The Measure of a Messenger: Meter as potential avenue of differentiation

Charles Pletcher[1,*]

[1]Tufts University, Department of Classical Studies, 10 Upper Campus Road, Medford, Massachusetts 02155, United States

Abstract

This paper searches for a metrical definition of messenger-type speech by way of arguing that such speeches (*Botenberichte*) in the ancient tragedies of Aeschylus, Sophocles, and Euripides have not received the computational attention that they deserve, but also that such attention is impossible given the current dearth of data for Ancient Greek tragedy. I first draw attention to this gap by surveying traditional literary approaches to the genre, showing how the definition of "messenger speech" is already problematic from a qualitative perspective. Next, I present two quantitative approaches to the meter of messenger speeches, neither of which manages to identify any distinguishing characteristics. In conclusion, I offer a brief discussion of these preliminary findings and potential avenues for further inquiry.

Keywords

tragedy, messenger speeches, iambic trimeter, metrical resolution

1. Introduction

This paper asks whether we can detect a "metrical signature" for messengers' speeches[1] in ancient Athenian tragedy. It pursues this line of inquiry through quantitative analyses of the meter in the three canonical tragedians (Aeschylus, Sophocles, and Euripides). In so doing, this paper sheds new light on questions that have previously been addressed mainly from qualitative angles, challenging the prevailing view that we can easily distinguish messenger-type speech from other iambic trimeters (the main spoken meter of tragedy).

In the first pages of *Narrative in Drama: The Art of the Euripidean Messenger-Speech*, Irene J. F. de Jong offers the following definition of (Euripidean) messenger-speeches: "The Euripidean messenger-speech or ἀγγελικὴ ῥῆσις is a long, continuous speech in which a messenger reports events that have taken place off-stage. The content of this speech is a narrative, more particularly a first-person narrative…" [1, 1–2]. She goes on to argue that Euripides makes signficantly greater use of first-person narrative than Sophocles and Aeschylus, citing [1, 2 n. 4]

- Aeshylus, *Persians* 480–514
- Aeschylus, *Agamemnon* 636–80
- Sophocles, *Antigone* 407–40, 1192–1243
- Sophocles, *Oedipus Tyrannus* 1237–85
- Sophocles, *Oedipus Colonus* 1586–1666

As the only examples of first-person narrative outside of Euripides' oeuvre. Although de Jong takes great pains (and around sixty pages) establishing the first-person focalization of Euripides' messenger-speeches, a cursory glance at, for example, the Tutor's speech in Sophocles' *Electra* (680–763) casts

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*Corresponding author.
charles.pletcher@tufts.edu (C. Pletcher)
https://github.com/pletcher (C. Pletcher)
0000-0003-2734-5511 (C. Pletcher)

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[1]Throughout this paper, I use "messenger-type speech" to refer to the category of speeches under investigation. But "messenger" describes a heterogenous class of individuals who do not necessarily conform to a single role or character type. I thus also use the term "news-bringer" when I mean to include characters who deliver news but who are not generally considered "messengers." Further, the analyses in this paper

enough doubt on de Jong's criteria that it raises further questions about not only Euripidean messenger-speech exceptionalism but messenger-speech as a category in its own right. [2] In this paper, I present two computational null results that cast further doubt on messengers' exceptionalism. By showing how ordinary messengers' speeches are, I mean both to highlight their dramatic—as opposed to narrative—place and to demonstrate how their performances contribute to tragedy's dramatic negotiations for truth.[3]

The paper begins with a brief and necessarily incomplete overview of the massive (and growing) bibliography on messengers and their speeches. I then discuss the methodologies and experiments by which this paper's results were obtained, beginning with the library that was used for scanning iambic trimeters, the spoken meter of Greek tragedy and thus the meter for most messenger-type (Euripides singing Phrygian in *Orestes* marking a notable exception). The first two experiments examine two interrelated ways in which meter might be supposed to distinguish messengers' speeches from other characters' speech, but which instead show that meter does not sufficiently distinguish messengers from other tragic speakers. The third experiment uses TF–IDF and 4-character-grams to cluster the approximately 17,000 lines of iambic trimeter in tragedy with all of the speeches from the Homeric corpus, obtained through the Digital Initiative for Classics: Epic Speeches (DICES) API. These experiments and their results are far from exhaustive, but they offer a new way of understanding the place of messengers in drama—a place, moreover, that appears no less "dramatic" than their traditionally less "narrative" peers.

2. A brief history of *Botenberichte*

The usual history of (so-called) "messenger speeches" in drama begins with Homer: the ancient bard's narrative *enargeia*, to borrow a term from Pseudo-Longinus, is said to animate the off-stage events that news-bringers narrate. De Jong has perhaps most famously applied a narratological lens to both Homer and tragic messengers, followed a few years later by Barbara Goward and James Barrett with their own seminal works on messenger speeches in 2002 [4, 1, 5, 6]. De Jong championed the view that messengers are not featureless conduits for the dramatist's voice but are instead focalizers with bias and selective memory. Barrett then qualified this view by noting that messengers' narrative effectiveness is inversely proportional to their stage presence: "The messenger's own body," he writes, "is implicitly—and explicitly, on occasion—a hindrance, inasmuch as his success as a messenger depends upon his acquiring the invulnerability granted by freedom from bodily constraints" [6, 100]. More recently, Margaret Dickin showed how the roles of messengers in tragedy increased over time and became "vehicles for star-power," while others, like Felix Budelmann and Evert van Emde Boas, explored how messengers direct the audience's affective energies [7, 8]. Florence Yoon's interventions have further disrupted the "category" of messengers by revealing the mistakes of painting with too broad a brush: *angeloi*, messengers proper, and *kerykes*, "heralds," perform different functions in tragedy [9].

The chief difficulty running through much of the literature on messenger speeches has to do with defining who, precisely, is a news-bringer[4] and what, precisely, is a messenger speech. Part of the difficulty arises from the fact that named characters at times appear to deliver speeches that resemble messenger speeches: nearly every named character in Sophocles' *Trachiniae* does so, as Sir Richard Jebb noted over a century ago [10]; and even Clytaemestra's famous "Beacon Speech" in Aeschylus' *Agamemnon* resembles speeches of news-bringers in its subject-matter. Thus, some scholars insist on anonymity as a prime criterion for the deliverer of a messenger speech, but such an insistance runs

[2] The Tutor's speech is rightly considered an exemplary "messenger-type speech" in its own right (so [2] and [3], among others), and it is largely for this reason that its apparent violation of Euripidean messenger-speech exceptionalism raises other questions.

[3] I use "messenger-speech" when referring to de Jong's concept, and "messenger's speech," "messengers' speeches," or "messenger-type speech" when referring to the general category of dramatic monologue.

[4] Along with ἄγγελοι, "messengers," I also count κήρυκες, "heralds," and various other unnamed deliverers of information. For the purposes of the computational studies that follow, however, I divide these speakers into different classes, described below. As I hope to show, any metrical distinctions between classes of characters in tragedy are artificial at best.

the risk of obscuring the similarities that so-called "messenger speeches" have with respect to other monologues in drama. These similarities matter, I argue, because they highlight the various strategies that drama uses to interrogate the fraught and problematic delivery of information that the recipients cannot verify. The tragedians, to state the matter bluntly, offer some of the earliest extended encounters with "fake news."

Yet despite the steady stream of innovative approaches to messenger speeches in tragedy, and despite Simon Perris' call for a "usable, robust, formal working definition" of messenger speech, no clear qualitative definition has emerged.[11] On the quantitative side, there have been few attempts to address the various classes of news-bringers from a computational angle. Where such examinations have occurred, they have, to my knowledge, taken place in the context of broader computational studies of drama, such as the work of Julia Jennifer Beine, Frank Fischer, and Viktor Illmer presented at the Digital Humanities Conference in 2024; the examination of the chorus by María Teresa Santa María Fernández and Monika Dabrowska; or the early work on social network theory and Greek tragedy by Jeff Rydberg-Cox [12, 13, 14].

In this paper, I undertake two quantitative approaches to meter in tragedy, with a focus on whether messengers and their ilk exhibit a discernible metrical signature by which they might be differentiated from other speakers. I investigate two specific phenomena: (1) **the number of resolutions per line of each character's speech**, that is, the number of times that a single long syllable in an iamb is replaced by two short syllables; and (2) **the placement of resolutions in characters' trimeters**, which has not, to my knowledge, been used as a way to distinguish between classes of speakers, although the *frequency* of resolution has been used for dating the plays of Euripides and Sophocles.

At the risk of discouraging the reader from continuing, both experiments turn up null or nearly results, with extremely weak correlation at best: the analyzed features are insufficient to distinguish messengers from other classes of tragic characters. Yet the reasons for this lack of differentation can still lead to a more nuanced understanding of the *dramatic* place of messengers in tragedy, as opposed to their relegation to "*narrative* in drama." By exploring the ways that we can measure a character's "messenger-type-ness," or the extent to which they perform a "messenger function" (to borrow a term from C. W. Marshall [2]), we shed new light on this often-misunderstood class of dramatic actors and thus better understand how they reflect their culture's relationship to secondhand information.

3. Data preparation

Data were prepared using TEI XML files from DraCor and the Perseus Project. For determining scansion, a rules-based library was built in the Elixir programming language. Although the deviation from Python might seem unusual, Elixir has the benefit of platform stability—anecdotally, Python projects break approximately every six months, while I have been able to maintain Elixir-based experiments for nearly a decade without major interruption— as well as first-class support for multi-threaded and parallel computing, techniques that will be beneficial for larger corpora. This early programming-language decision will mean straightforward parallelization as/if the corpus grows. It also means that these tools are reusable for larger projects, even if the developers of those projects are not familiar with Elixir: they need only install the Elixir runtime and run the pre-written processing scripts to output their own scansion analysis.

Experiments were run in Livebook, an Elixir-based corollary to Jupyter notebooks. The relevant Livebooks and supporting files are available upon request. (They are currently hosted on a platform other than GitHub and so cannot be anonymized.)

For the sake of efficiency, the rules-based scansion parser was initially built with the assistance of Claude Code. While some of that code remains in the final implementation, the author found several major bugs that were leading to under-counting the number of trimeters in characters' speech—trimeter should make up about 60% of tragic dialogue, but the "vibe-coded" implementation of the metrical analyser only captured 40%. Thus, the author rewrote much of the code to fix these errors, and the library now reports valid trimeters as 57% of lines scanned. The author verified the scansion results by

examining key passages from each of the three playwrights, and the author wrote all of the experiments and conducted the analysis without AI assistance.

The scansion parser first syllabifies a text and then parses line by line to determine whether the line can scan as valid iambic trimeter. Out of 44,482 lines of tragedy in the corpus, 26,433 scan as valid trimeters; 81 were skipped due to parsing errors. The corpus of iambic trimeters thus represents about 57.17% of the lines of tragedy in the corpus, matching the rough division between odes and episodes in most plays, which hovers around 60%.

For all of the experiments, each speaker was labeled with one of eight "speaker classes": (1) messenger, (2) herald, (3) chorus, (4) royalty, (5) divinity, (6) slave, (7) prophet, or (8) other. These labels were first applied automatically to unambiguous characters—all Ἄγγελοι were labeled "messenger," for example. Subsequently, a list of ambiguous characters was generated and labeled by hand.

The definition of messenger for the purposes of labeling was deliberately strict. Although, as noted above, the category of messenger-type speech has notoriously fuzzy boundaries, the labeling for these experiments followed character naming conventions exclusively. We should note that this labeling method is still artificial: while we cannot know specifics about when line assignemnt labels entered the manuscript tradition, our earliest manuscripts include only paragraphoi to signal a change of speaker, but they do not label the speakers specifically.[5] Nevertheless, it seemed best to use what are now the canonical labels for speaker class, rather than to rely on necessarily subjective judgment while attempting to establish quantitative parameters for identifying messenger-type speech.

4. Parsing iambic trimeter

The rules-based parser works as follows (visualized also in the accompanying above): (1) First, the raw Greek text is read to determine whether it is Beta Code or polytonic Greek. These experiments work exclusive with polytonic Greek, so this step is essentially a noop. (2) Second, the raw text is tokenized by grapheme (letters with diacritical markings attached). (3) Subsequently, the graphemes are grouped by syllables, accounting for diphthongs, elision, and double consonants. Compounds such as εἰσ-φέρω are not explicitly handled, as doing so would add out-of-scope morphological analysis to this scanner with minimal improvement on the scansion analysis: for example, in the case of εἰσ-φέρω, εἰ- will always scan long, and the epsilon in φέρω will always scan short, whether the compound is grouped correctly as εἰσ-φέρω or incorrectly as εἰ-σφέρω. (4) Syllables are classified as short, long, anceps, or unknown according to standard rules of Ancient Greek scansion. According to the usual convention in tragic verse, *muta cum liquida* (a stop consonant followed by a lambda or a rho) does not result in the usual lengthening that applies to other short vowels followed by two consonants. Other double-consonant clusters will cause preceding short vowels to be scanned long by this scanner.[6] (5) The scanned syllables are mapped onto the iambic trimeter schema of three dipodies (in its most basic form: x — u — | x — u — | x — u —). (6) Features such as resolution and dactylic events are labeled according to the flow chart below.

[5] See [15].

[6] Unknown vowel quantities are logged in a dictionary for later manual annotation. This annotation process has not yet been completed.

(1) raw text
↓
(2) grapheme tokenization
↓
(3) syllabification
↓
(4) quantification (labeling of short, long, and anceps syllables)
↓
(5) parsing (mapping to iambic trimeter)
↓
(6) feature extraction (label resolutions and dactylic events)

Dactylic events refer specifically to the possibility of a long anceps followed by a resolved iambic "up-beat," i.e., the familiar epic — u u (long short short). A hypothesis concerning the "epic-ness" of messengers in tragedy might suppose that they make greater use of dactyls than other classes of speakers. This hypothesis turns out false, however.

5. Resolution count

We begin by counting resolutions in iambic trimeters. The frequency of resolution offers a sense of how often each character class adds more syllables than are strictly necessary to a line of iambic trimeter. The strength of the relationship between speaker class and resolution count was tested using [Harald] Cramér's φ_c (also known as Cramér's V). As Cramér writes, "The correlation coefficient and the correlation ratio both server to characterize, in the sense explained above, the 'degree of dependence' between two variables.... $\varphi^2 = 0$ when and only when the variables are independent." [16, 282] Cramér goes on to explain that φ_c will range between 0 (no correlation) and 1(full corrleation). Thus, by measuring Cramér's φ_c for speaker class and number of resolutions, we can determine whether the two are related and the strength of that relation.

For 8 speaker classes, the correlation with resolution count was 0.0437—meaning a miniscule connection between the two variables. The figure below shows the relative frequency of resolutions per total lines by speaker class.

speaker class	**resolutions**	**total lines**	**rel. freq. resolved**
prophet	89	350	0.2543
other	270	1478	0.1827
chorus	289	1685	0.1715
messenger	383	2336	0.1640
royalty	2445	16205	0.1509
divine	265	1927	0.1375
slave	83	893	0.0929
herald	1	665	0.0859

6. Resolution placement

As with total resolutions, resolution placement has a negligible positive correlation with speaker class, peaking in positions 6 (0.0924) and 2 (0.0622). These correlations are meaningful, but they are too small with our limited data to suggest predictive value.

position	**Cramér's φ_c**
2	0.0622
4	0.0468
6	0.0924
8	0.0583
10	0.0184

Indeed, we can see small but meaningful differences in the absolute and relative frequencies for each resolution for every position but 10. (See full results at »REDACTED FOR ANONYMITY«) These differences, however, hardly provide any descriptive value. The relative frequencies of resolution are so low for each position—generally under 10%— that it is difficult if not impossible to infer a metrical signature for any class of speaker from data drawn from the extant plays.

Below are visualized the relative frequencies of resolution by position and by tragedian. The breakdown by tragedian is necessary because of the state of the extant corpus: 19 (counting, perhaps contraversially, *Rhesus*) of Euripides' plays survive, compared to 7 each for Aeschylus and Sophocles (7.5 for Sophocles if we count *Trackers*). Relative frequencies are visualized both as a proportion of each speaker class's resolution placement and as a grouped bar chart across speaker classes.

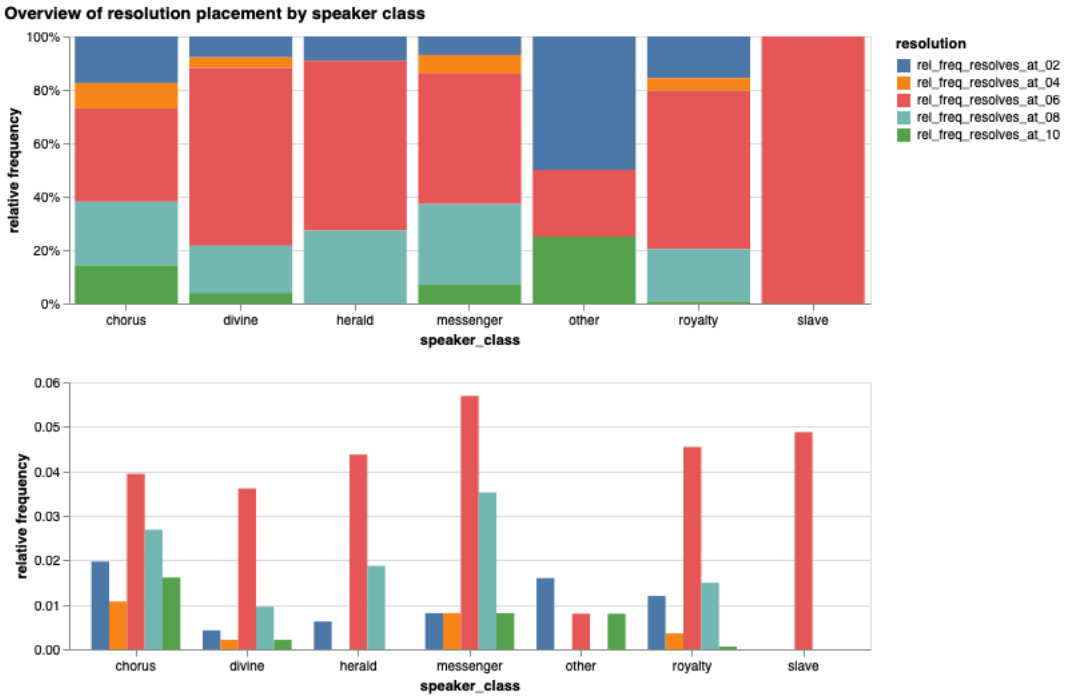


Figure 1: Relative frequencies by resolution placement for Aeschylus

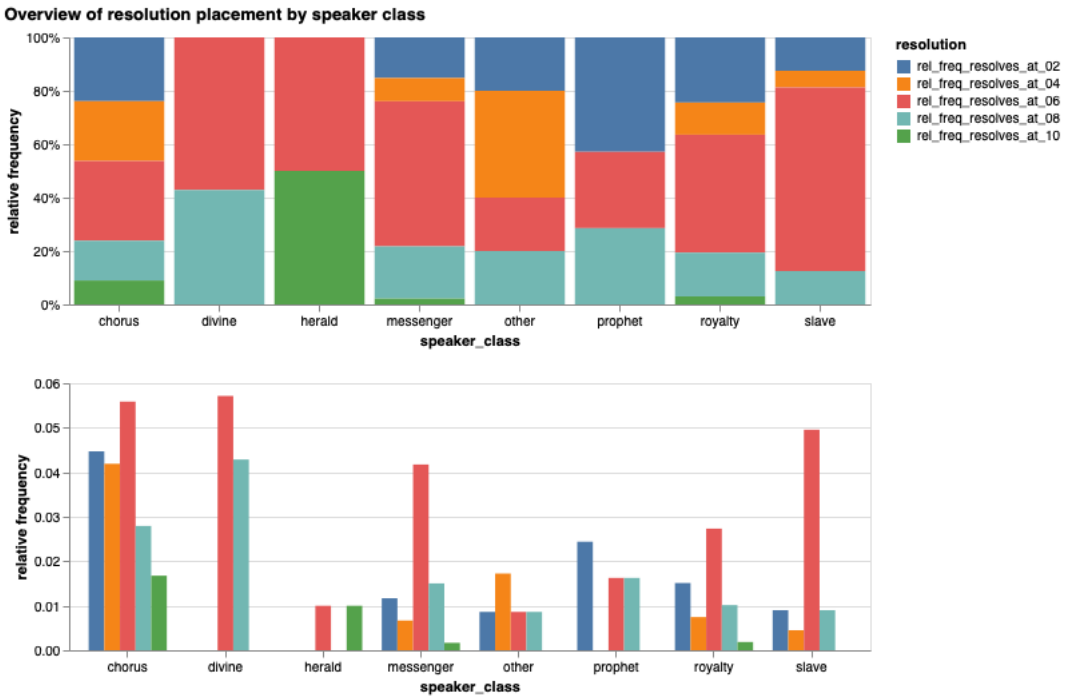
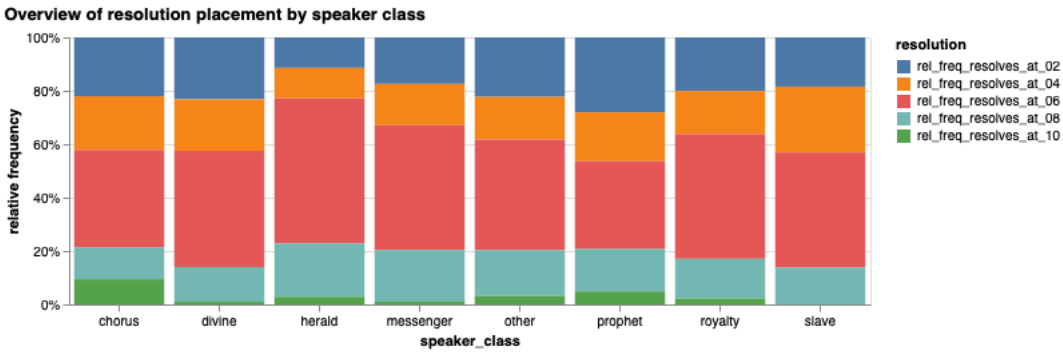


Figure 2: Relative frequencies by resolution placement for Sophocles



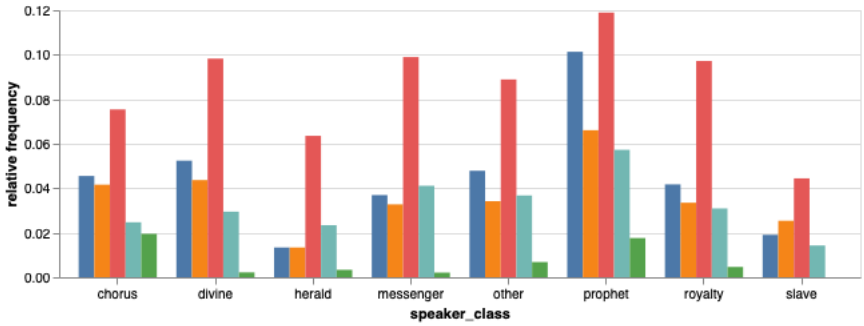


Figure 3: Relative frequencies by resolution placement for Euripides

As is readily visible in these figures, resolution count varies significantly by dramatist, with Euripides' maximum relative frequency (11.9% for prophets) more than doubling the maxima of Sophocles (5.7% for divinities) and Aeschylus (5.7% for messengers). Aeschylus and Sophocles exhibit more variable resolution placement among speaker classes than Euripides, but this observation is likely due to the smaller sample size for these tragedians: the data include Aeschylus' 7 canonical plays (including *Prometheus Bound*) and Sophocles' 7 plays and the substantial fragments of his satyr play (*Trackers*); but the data also include all 17 of Euripides' tragedies and 2 of his satyr plays(*Cyclops* and *Alcestis*), meaning that we have significantly more information for his plays than for the plays of the other two tragedians. The marked differences in distributions between the three tragedians' resolution placements supports separating our analyses; but at the same time, the smaller sample sizes for Aeschylus and Sophocles limit our ability to draw inferences from the resolution distributions.

7. Discussion

These experiments show that there *are* detectable metrical differences among speaker classes in tragedy, perhaps especially among messengers. However, these experiments also point up the impossibility of making meaningful classifications based on the available data: we could not for instance, take an unassigned line from the tragic fragments and determine whether it was a messenger's or not. These null quantitative results should not be understood as contradicting the more literary analyses of de Jong and others. This paper's results only suggest that meter fails as a distinguishing feature of messenger-type speech; but other features, such as lexis and syntax, might still reveal key differences between messengers and other types of characters.

These differences matter because messengers model how audiences understand, interpret, and evaulate second-hand news and the people who bring it to them. But in the case of this paper, the *lack* of apparent metrical differences matters, too. Messengers, according to interpreters like Barrett, de Jong, and Dickin, acquire authority from the way that they deliver their message, and it has long been a truism that messengers speak with "Homeric diction." But what is "Homeric diction," and how might

we identify it, especially when messengers' speeches appear to use the same dialect as other spoken sections of tragedy.

Part of the difficulty here lies in the limited data available for quantitative analyses of Athenian drama. As noted above, the entire corpus under analysis here consists of 44,482 lines, of which we analyzed 18,968 trimeters. Over half of those lines belong to Euripides' plays; the other half skews slightly Sophoclean. But this relatively small dataset and its unequal distribution mean that quantitative analyses are necessarily limited in scope. As Quinn Dombrowski and Patrick Burns note, languages other than English are at a severe disadvantage when it comes to computational study.[17] Although quantitative study of drama has been conducted successfully by [18] and others, those studies were carried out on larger corpora with a greater concentration of manual annotations than are currently available for Ancient Greek.

Especially for poetic texts like tragedy, the state of the art lemmatizers and morphological tools—which have been trained overwhelmingly on prose treebanks—do not perform sufficiently well for nuanced analysis. This limitation motivated the present study's focus on meter using a relatively naïve rules-based parser on a relatively straightforward meter. (By avoiding anapests and odes, the metrical parser could avoid many of the pitfalls that would require extensive training and manual annotation/intervention.) Yet even with these self-imposed guardrails, the present study could not marshall sufficient data to detect any metrical signature on the part of character-type speech.

Under-resourced languages must find ways to overcome these barriers if computational scholars are to be seen on equal footing with our English- (and Spanish-, French-, German-, Italian-, Chinese-, etc.) language–focused colleagues. As a start, we need to recognize the labor that goes into annotating textual sources. The Text Encoding Initiative is something as a barrier here, as it focuses on physical texts rather than the abstract textual models needed by editions of drama.[7]

But we must also find ways of posing questions the answers to which are not limited by the limitations of available data. These methodological problems deserve scholarly attention, as a richer understanding of under-resourced languages—ancient or modern—leads to a richer appreciation even of well-resourced fields of inquiry.

8. Future directions

Even if we cannot use meter to distinguish messenger-type from other kinds of speech in tragedy, "Homeric diction" might still be a useful heuristic for syntax and lexis. To this end, future work will focus on possible lexical distinctions among different types of speakers in tragedy. Bag-of-words models offer some reprieve from the limitations of line-oriented studies (like the present examinations of meter), but successful bag-of-words approaches depend either on sub-word tokenization or better lemmatization than is currently available. Studies of syntax are likewise limited by the accuracy of current morphological tools on poetry.

In a separate vein, future work might focus on network analysis, which would require additional manual annotations of existing corpora. Such annotations could be accomplished by a small team of researchers in a reasonably short amount of time—for instance, by adding "toWhom" attributes to dialogue tags in TEI-encoded dramas—but could yield interesting results about internal audiences and attention in the extant plays.

Above all, as this paper has shown, future work should address the methodological gap between the state-of-the-art models—which focus almost exclusively on English—and under-resourced langauges like Ancient Greek. By developing methodologies that can account for the limitations of small corpora, we can direct efforts to annotate and supplement those corpora while simultaneously conducting meaningful research.

[7] See [19].

9. Declaration on Generative AI

The author built the metrical parsing library with the help of Anthropic's Claude Code—and subsequently rewrote most of it. No AI was used in the calculations or analysis that ultimately make up the bulk of this paper.

References

[1] I. J. F. de Jong, Narrative in Drama: The Art of the Euripidean Messenger Speech, Brill, 1991.

[2] C. W. Marshall, How to Write a Messenger Speech (Sophocles' Electra 680–763), Bulletin of the Institute of Classical Studies. Supplement (2006) 203–221.

[3] J. H. Kells, Sophocles: Electra, Cambridge University Press, 1973.

[4] I. J. F. de Jong, Narrators and Focalizers: The Presentation of the Story in the Iliad, Bristol Classical Paperbacks, 2. ed., impr ed., Bristol Classical Press, 1989.

[5] B. Goward, Telling Tragedy, Duckworth, 1999.

[6] J. Barrett, Staged Narrative: Poetics and the Messenger in Greek Tragedy, University of California Press, 2002.

[7] M. Dickin, A Vehicle for Performance: Acting the Messenger in Greek Tragedy, University Press of America, 2009.

[8] F. Budelmann, E. van Emde Boas, Attending to Tragic Messenger Speeches, in: J. Grethlein, L. Huitink, A. Tagliabue (Eds.), Experience, Narrative, and Criticism in Ancient Greece : Under the Spell of Stories, Oxford University Press, 2020, pp. 59–80.

[9] F. Yoon, Heralds and Messengers: Character Identity and Function in Greek Tragedy, Classical Philology 117 (2022) 364–385. URL: https://www.journals.uchicago.edu/doi/10.1086/718685. doi:10.1086/718685.

[10] R. C. Jebb, Sophocles: The Plays and Fragments, with Critical Notes, Commentary, and Translation in English Prose. Part V: The Trachiniae., Cambridge University Press, 1902.

[11] S. Perris, What Maketh the Messenger: Reportage in Greek Tragedy, in: A. Mackay (Ed.), Selected Proceedings, 2011. URL: www.ascs.org.au/news/ascs32/Perris.pdf.

[12] F. F. Julia Jennifer Beine, V. J. Illmer, Just the Type: Analysing Character Typology in Roman Comedy with RomDraCor (2024). URL: https://zenodo.org/doi/10.5281/zenodo.13286286. doi:10.5281/ZENODO.13286286.

[13] M. T. S. M. Fernández, M. Dabrowska, Análisis comparativo del coro como personaje en tres tragedias griega y tres dramas españoles del corpus dracor, Neophilologus 107 (2023) 389–412.

[14] J. Rydberg-Cox, Social Networks and the Language of Greek Tragedy, Journal of the Chicago Colloquium on Digital Humanities and Computer Science 1 (2011).

[15] D. J. Mastronarde, Text and Transmission, John Wiley and Sons, Ltd., 2016, pp. 9–26. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/9781119257530.ch2. doi:https://doi.org/10.1002/9781119257530.ch2. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119257530.ch2.

[16] H. Cramér, Mathematical Methods of Statistics (PMS-9), number 9 in Princeton Mathematical Series, Princeton University Press, 2016. doi:10.1515/9781400883868.

[17] Q. Dombrowski, P. J. Burns, Language Is Not a Default Setting: Countering DH's English Problem, Debates in Digital Humanities (2023) 295–304.

[18] J. J. Beine, How to Figure out a Schemer: Tracing Types of Roman Comedy in Classical Receptions through Digital Methods, in: D. Wouters, J. Bloemendal (Eds.), Transnational Encounters in Early Modern Drama, 1450–1750, 2025, pp. 308–331.

[19] T. Efer, Introducing NoXML for the Digital Humanities, Informatik (2017) 1161–1166. doi:10.18420/IN2017_117.